

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		$\text{CH}_3\text{COOH} + \text{NH}_2\text{CONH}_2 \rightarrow \text{CH}_3\text{CONH}_2 + \text{CO}_2 + \text{NH}_3$		1		1		
		(ii)		it contains a <u>nitrogen</u> atom that has a lone pair of electrons / is a proton acceptor		1		1		
		(iii)		the C=O absorption at 1650-1750 cm^{-1} decreases / the N—H absorption at 3300-3500 cm^{-1} decreases (1) the C≡N absorption at 2100-2250 cm^{-1} increases (1)		2		2		
		(iv)		award (1) for any of following - the benzene ring / negative electron cloud is not (easily) susceptible to attack by nucleophiles - the benzene ring / negative electron cloud is usually attacked by electrophiles - the C—Cl bond in chlorobenzene is stronger than the C—Cl aliphatic bond			1	1		
				Question 10 total	2	10	1	13	5	1